Sharing the Gains of Aggregation: Cooperative Imbalance Cost Allocation

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***Abstract*—A *facilitator* is an intermediary that offers renewable producers and consumers fixed-price contracts and, acting as their balance responsible party, manages the residual imbalances in the market. Pooling imperfectly correlated residuals nets consumers' imbalances and reduces the portfolio's total imbalance cost, but raises an allocation question: how should these imbalance cost savings be divided among heterogeneous consumers? We formulate this as a cooperative game, the *imbalance netting game*, and compare six allocation mechanisms under a two-price imbalance settlement. Our central finding is that aggregation invariably reduces total costs regardless of mechanism; the mechanisms differ in how they redistribute the savings, their computational requirements, and whether they satisfy the core and group rationality conditions. We establish two analytical results: a necessary and sufficient condition for budget balance in the marginal cost contribution mechanism, and a condition for when the Gately point is well-defined. On Danish 2024 data, flat-rate socialization violates group rationality for some consumers, while full individualization recovers total costs but transfers cost variability to consumers. Partial individualization maintains group rationality and recovers costs while limiting consumer cost exposure.**

***Index Terms*—Cost allocation, cooperative game theory, imbalance netting game, aggregation, balancing market**

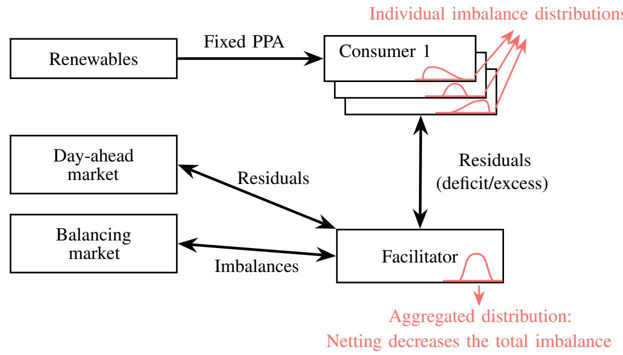

Fig. 1: System overview. Each consumer holds a fixed-price Power Purchase Agreement (PPA), which is a fixed share of renewable production. The facilitator acts as the Balance Responsible Party (BRP) for the resulting residual net positions, bidding to cover expected shortfalls and offering expected surplus in the day-ahead market, then settling deviations in the balancing market. Pooling imperfectly correlated residuals reduces the portfolio's imbalance. Black arrows are power transactions.

I. Introduction

As electricity markets liberalize and the share of variable renewable generation increases, new intermediaries are emerging to shield consumers and renewable producers from market and balancing risk. We refer to such an intermediary as a *facilitator*. The facilitator negotiates Power Purchase Agreement (PPA)s between renewable producers and consumers and manages the *residual*, i.e., the part of each consumer's net position not covered by its PPA, which is uncertain because both production and consumption are. A concrete example, and the motivation for this work, is the Danish supplier Reel[1], which gives a portfolio of producers and consumers a fixed price and manages the residual balancing risk of the whole portfolio.

The facilitator therefore has two roles: brokering the PPAs and managing the residual. In this paper, the PPA contracts are taken as fixed and given, and we focus on the allocation of the residual imbalance cost. As illustrated in Fig. 1, each consumer holds a fixed-price PPA covering a share of renewable production, while the remaining net position must be traded in the market. Because both demand and production are uncertain, this position carries forecast risk. Acting as the Balance Responsible Parties (BRP) for the portfolio, the facilitator bids the aggregated residual in the day-ahead market, settles the resulting deviations in the balancing market, and prices the *imbalance cost* back to its consumers. By managing this residual risk, it extends the bankability of PPAs to smaller consumers and projects, and offers consumers a hedge against balancing risk they may not manage on their own.

The consumer's bill therefore has a simple structure. For the energy component, excluding taxes and grid tariffs, each consumer pays a fixed PPA price, a fixed service fee to the facilitator, day-ahead market prices for any net residual consumption or generation, and a share of the portfolio's imbalance cost. The first two are set in advance and do not change. Any residual volume is settled at the day-ahead market price. The imbalance share is charged ex-ante as an expected amount and then reconciled against realized cost after settlement, to an extent the facilitator chooses through its degree of cost individualization: from fully socialized, where the expected charge stands and the facilitator bears the deviation, to fully individualized, where each consumer ultimately pays its realized cost. Of these components, only the imbalance share and its division among consumers are studied here. The PPA price and the service fee are taken as given.

The economic value of this service arises from a demand-side *portfolio effect*. The consumers' residual positions are stochastic and imperfectly correlated, so their forecast errors partly offset when pooled. An imbalance from one consumer may be netted by an opposite imbalance from another, reducing the portfolio's relative imbalance volume and, under suitable pricing, its total imbalance cost. Aggregation can thus create a genuine cost saving relative to the sum of the

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[1]https://reel.energy

consumers' standalone imbalance costs.

The central question is how this saving should be allocated. A fully socialized scheme, in which all consumers pay the same average imbalance cost per unit of demand, is simple but ignores heterogeneity in forecastability, size, and contribution to portfolio imbalance. Charging every consumer its standalone imbalance cost, conversely, ignores the netting value that exists only through aggregation. Neither extreme is satisfactory for a heterogeneous portfolio, which motivates the research question of this paper: *how should the imbalance cost in an aggregated BRP portfolio be allocated among consumers so as to be fair, stable, budget balanced, and practically implementable?*

We study this question for inelastic consumers in the day-ahead and balancing markets, so that a consumer's imbalance stems from the deviation between its contracted and realized residual volume rather than from any dispatch decision. Imbalances are settled under an asymmetric two-price imbalance settlement: deviations that help balance the power system are settled at the day-ahead price, while those that aggravate the system imbalance are settled at a less favorable imbalance price. We adopt this setting because it quantifies directly the value of reducing imbalance volume as an expected cost saving[2].

The value of aggregation under uncertainty is well established. As renewable penetration grows, market participants face higher uncertainty [3], and aggregating geographically or technologically diverse resources can reduce aggregate variability [4], [5]. As virtual power plants and other multi-owner aggregations become prevalent [6], allocating the gains of aggregation becomes a central market design problem, for which cooperative game theory offers a natural framework. It has been applied to energy communities [7], shared storage access [8], and alternatives to marginal pricing in wholesale markets [9], and Baeyens et al. [10] use it to share the gains of aggregating wind producers in forward markets.

Uncertainty is increasingly a demand-side phenomenon as well: the growth of distributed generation and intermittent self-sufficiency makes load-related risk a major driver of retailer and BRP exposure [11]. This motivates studying aggregation of consumers' residual demand. Closest to our setting, Guo et al. [12] aggregate co-located data centers under a marginal price allocation, and Valencia Zuluaga and Oren [13] show that this allocation preserves fairness in peer-to-peer markets. These works establish the marginal price allocation as an important benchmark, but do not compare it systematically with other cooperative mechanisms on a facilitator's portfolio, nor examine how the degree of cost individualization affects consumer-level payments, stability, and risk sharing.

This paper makes four distinct contributions. First, we cast imbalance cost allocation in a facilitator's portfolio as a cooperative game, the *imbalance netting game*, where coalition costs are determined by the two-price imbalance settlement. This framework makes fairness and stability operationally precise through the core and group rationality conditions. Second, we systematically compare six allocation mechanisms (Shapley value, marginal cost contribution, Vickrey-Clarke-Groves (VCG), Nucleolus, marginal price allocation, and Gately point) across four properties: computational tractability, budget balance, group rationality, and additivity. We identify which mechanisms remain practical as portfolio size grows. Third, we derive two novel theoretical results: (i) a necessary and sufficient condition for the marginal cost contribution mechanism to be budget balanced, showing that when violated, it systematically under-recovers costs and leaves the facilitator in budget deficit. (ii), a condition for when the Gately point is well-defined, with the unique core allocation that applies when it is not. Fourth, on a Danish case study of nineteen consumers sharing a photovoltaic PPA with 2024 data, we quantify how the mechanisms redistribute aggregation gains. We show that partial individualization balances fairness, consumer risk exposure, and facilitator incentive alignment.

The remainder of this paper is organized as follows. Section II formulates the imbalance netting game and the pipeline for computing imbalance costs. Section III presents the six allocation mechanisms. Section IV provides the analytical results. Section V applies the framework to the case study. Section VI concludes.

II. The Imbalance Netting Game

We model the cost allocation as a cooperative game [14], in which a set of $|N|$ consumers of the facilitator can form binding coalitions, that is, subsets $S \subseteq N$ that cooperate to reduce the total procurement cost borne by the group; the full set $N = \{1, 2, \ldots, |N|\}$ is the *grand coalition*. The procurement cost has two parts: the cost of energy bought in the day-ahead market, and the imbalance cost settled in the balancing stage. For instance, a facilitator with three consumers has $2^3 = 8$ possible coalitions, including the empty coalition and the grand coalition. We write $N \setminus \{i\}$ for the coalition of all consumers except i.

Remark 1. *We consider a business model for the facilitator in which consumers pay the day-ahead price for electricity plus a balancing fee, so the day-ahead procurement cost is always covered by the consumer and only the price spread between the day-ahead and balancing markets is exposed to the facilitator. The game therefore reduces to allocating the imbalance cost alone, the sole cost the facilitator must split among the consumers.*

Each coalition is assigned a value by a characteristic function $c : 2^N \to \mathbb{R}$, and here $c(S)$ is the imbalance cost incurred by coalition S, so that $c(S) \geq 0$ for every coalition S. Of particular interest is the *grand-coalition cost* $c(N)$, the total imbalance cost of the fully aggregated portfolio. We now cast the allocation of imbalance costs as a cooperative game and define the mechanisms that solve it.

[2] Under a symmetric one-price scheme, the same aggregation problem can be posed, but the primary benefit of pooling is reduced income volatility rather than reduced imbalance cost. The two-price setting is also relevant given regulatory developments: while symmetric imbalance pricing is gradually replacing asymmetric schemes across parts of Europe under regulatory requirements [1], the Danish transmission system operator's upcoming move to *full cost balancing* points toward a more asymmetric "polluter pays" design [2], which reintroduces incentives to reduce imbalance volume.

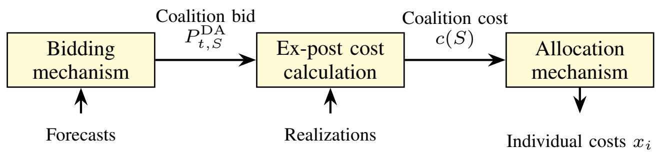

Fig. 2: Three-step pipeline of the imbalance netting game. Individual forecasts of renewable production and consumption, made by the facilitator, feed the bidding step to produce day-ahead quantity bids $P_{t,S}^{\mathrm{DA}}$ for each coalition $S \subseteq N$. Realizations feed the ex-post cost calculation that defines the characteristic function $c(S)$, which shows the imbalance cost for the coalition S. The allocation mechanism divides the grand-coalition imbalance cost $c(N)$ into individual consumer costs x_i.

Definition 1 (Imbalance netting game)**.** *The imbalance netting game is the cooperative game (N, c) whose players are the facilitator's consumers and whose characteristic function $c(S)$ equals the imbalance cost incurred by coalition S's aggregated residual under the two-price imbalance settlement. The value is not given a priori; it is computed for each coalition through the pipeline of Fig. 2: a group quantity bid is formed (Section II-A) and settled ex-post to obtain the grand-coalition cost that defines c (Section II-B), after which such a cost is divided among consumers by an allocation mechanism (Section III).*

The first two steps construct the game and the third solves it. We describe each in turn.

Remark 2. *The facilitator generates all forecasts centrally, including renewable production and each consumer's consumption. Net-demand forecasts are then calculated as consumption minus its PPA-based share of renewable production. As a result, consumers cannot influence the game by misreporting their own forecasts.*

A. Bidding Mechanism

We bid each consumer's residual individually at the newsvendor-optimal quantile that minimizes its expected imbalance cost, and the facilitator submits the aggregate as the coalition bid. We use a probabilistic bidding strategy similar to the one proposed by [15], under which the optimal day-ahead quantity bid (in MW) for consumer i in hour t is

$$P_{t,i}^{\mathrm{DA}} = \left(G_t^{\mathrm{E}_i}\right)^{-1}(\alpha_t), \qquad \forall t \in \mathcal{T},\ \forall i \in \mathcal{N}, \tag{1a}$$

$$\alpha_t = \frac{\pi_t^-}{\pi_t^+ + \pi_t^-}, \qquad \forall t \in \mathcal{T}, \tag{1b}$$

where $\mathcal{T} = \{1, 2, \ldots, |\mathcal{T}|\}$ is the set of hourly settlement periods over the study horizon, $G_t^{\mathrm{E}_i}(\cdot)$ is the cumulative distribution of net demand for consumer i, and π_t^+ and π_t^- are the penalties for positive (surplus) and negative (deficit) deviation. This is a newsvendor problem: α_t is the critical fractile that balances the cost of over- and under-bidding, and the optimal quantity bid is the corresponding quantile of the net-demand distribution. The distribution $G_t^{\mathrm{E}_i}(\cdot)$ is approximated by sampling historical consumer data, and π_t^+ and π_t^- by sampling historical price data. Because the critical fractile α_t depends only on prices and not on the coalition, coalition bids are formed by summing singleton bids, $P_{t,S}^{\mathrm{DA}} = \sum_{i \in S} P_{t,i}^{\mathrm{DA}}$, so that imbalances are additive across consumers by construction. Netting does not arise from the bid but from ex-post pricing: the coalition is settled on the sign of its aggregate imbalance $\Delta_{t,S}$, so offsetting individual deviations reduce the coalition's imbalance cost.

B. Ex-post Imbalance Cost Calculation

Recall that the coalition imbalance is additive, $\Delta_{t,S} = \sum_{i \in S} \Delta_{t,i}$. It is the coalition's day-ahead quantity bid $P_{t,S}^{\mathrm{DA}}$ plus its contracted share β_S of the realized portfolio renewable production P_t^{G}, minus its realized consumption $P_{t,S}^{\mathrm{D}}$:

$$\Delta_{t,S} = P_{t,S}^{\mathrm{DA}} + \beta_S P_t^{\mathrm{G}} - P_{t,S}^{\mathrm{D}}, \qquad \forall t \in \mathcal{T},\ \forall S \subseteq \mathcal{N}. \tag{2}$$

A positive imbalance $\Delta_{t,S} > 0$ means available energy exceeds consumption, a surplus, so the coalition is *long*; a negative imbalance $\Delta_{t,S} < 0$ means consumption exceeds available energy, a deficit, so the coalition is *short*. We write $\Delta_{t,N}$ for the *grand-coalition imbalance*.

Under the two-price imbalance settlement adopted here, a coalition's imbalance is priced by whether it *helps* or *harms* the imbalance of the entire power system: an imbalance that opposes the system imbalance (helping) is settled at the day-ahead price and earns nothing beyond it, while one that reinforces it (harming) is settled at a less favourable price. Specifically, if the power system is long, a power excess by a coalition is priced below the day-ahead price, meaning the coalition has lost the opportunity to sell that volume earlier at a higher price in the day-ahead market. Conversely, if the power system is short, a power deficit by a coalition is penalized above the day-ahead price. Under this scheme, the imbalance price can never improve on the day-ahead price, so there is no arbitrage opportunity between day-ahead and balancing markets and hence no incentive to create imbalance deliberately.

The bids are compared ex-post to the realized day-ahead and balancing prices, production, and demand, giving the total cost each coalition incurs in the day-ahead and balancing markets.

Under the two-price imbalance settlement, the *price spread* in hour t for coalition $S \subseteq N$ is

$$\lambda_{t,S} = \begin{cases} \lambda_t^{\mathrm{DA}} - \lambda_t^+, & \text{if } \Delta_{t,S} \le 0, \\ \lambda_t^{\mathrm{DA}} - \lambda_t^-, & \text{if } \Delta_{t,S} > 0, \end{cases} \qquad \forall t \in \mathcal{T},\ \forall S \subseteq N, \tag{3a}$$

where λ_t^+, λ_t^-, and λ_t^{DA} are the up-regulation, down-regulation, and day-ahead prices, respectively. The helping direction always settles at the day-ahead price, so $\lambda_t^+ = \lambda_t^{\mathrm{DA}}$ when the power system is long and $\lambda_t^- = \lambda_t^{\mathrm{DA}}$ when it is short. The spread $\lambda_{t,S}$ is therefore zero whenever the coalition helps the power system, and its product with $\Delta_{t,S}$ is strictly positive only when the coalition harms it. For the grand coalition we call $\lambda_{t,N}$ the *grand-coalition spread*.

The penalties π_t^+ and π_t^- in (1b) are the expected magnitudes of the two spread branches in (3a), taken over the historical price sample: a positive deviation (surplus) settles at the down-regulation spread and a negative deviation (deficit) at the up-regulation spread, so $\pi_t^+ = \mathbb{E}[\lambda_t^{\mathrm{DA}} - \lambda_t^-]$ and $\pi_t^- = \mathbb{E}[\lambda_t^+ - \lambda_t^{\mathrm{DA}}]$, both non-negative.

Remark 3. *As the facilitator is a price-taker, the balancing prices are fixed within each direction, so the spread a coalition faces changes only when its net position switches between helping and harming the power system, not when its imbalance volume grows.*

The characteristic function of coalition S is the sum over the horizon $\mathcal{T}$ of its imbalance priced at the spread,

$$c(S) = \sum_{t\in\mathcal{T}} \lambda_{t,S}\,\Delta_{t,S}, \qquad \forall S \subseteq N, \tag{4}$$

which is the imbalance cost of coalition S and satisfies $c(S) \geq 0$[3]. In summary, the two-price imbalance settlement gives four cases, where a coalition helps the power system when its imbalance opposes the system imbalance and harms it otherwise:

- **System long, coalition short** ($\Delta_{t,S} < 0$): the coalition helps, so $\lambda_t^{+} = \lambda_t^{\mathrm{DA}}$, the spread is $\lambda_{t,S} = 0$, and the coalition incurs no imbalance cost that hour.
- **System long, coalition long** ($\Delta_{t,S} > 0$): the coalition harms, is settled at the down-regulation price below day-ahead ($\lambda_t^{-} < \lambda_t^{\mathrm{DA}}$), therefore $\lambda_{t,S} > 0$, which incurs a positive imbalance cost.
- **System short, coalition long** ($\Delta_{t,S} > 0$): the coalition helps, so $\lambda_t^{-} = \lambda_t^{\mathrm{DA}}$, the spread is $\lambda_{t,S} = 0$, and the coalition incurs no imbalance cost that hour.
- **System short, coalition short** ($\Delta_{t,S} < 0$): the coalition harms, is settled at the up-regulation price above day-ahead ($\lambda_t^{+} > \lambda_t^{\mathrm{DA}}$), therefore $\lambda_{t,S} < 0$, and incurs a positive imbalance cost.

The game is a minimization of the imbalance cost $c(S)$. This completes the imbalance netting game (N, c).

C. Game Properties and Definitions

Two questions are essential: when is it beneficial for consumers to merge into larger coalitions, and how the resulting imbalance cost should be shared among them. The remainder of this section introduces the properties that answer these questions. The first question is one of coalition structure: whether combining coalitions ever destroys collective value.

Definition 2 (Sub-additivity)**.** *A game is sub-additive if* $c(S_1 \cup S_2) \leq c(S_1) + c(S_2)$ *for all disjoint* $S_1, S_2 \subseteq N$*.*

Sub-additivity guarantees that a merged coalition costs no more than its parts separately, so consumers are never worse off by merging. The imbalance netting game is sub-additive because pooling lets opposing imbalances cancel and reduce the imbalance cost, while aligned imbalances sum additively, so under the the price-taking assumption in Remark 3, the pooled imbalance cost never exceeds the sum of the consumers' standalone imbalance costs.

A stronger condition is concavity, which strengthens this from "merging never hurts" to "merging into larger coalitions is increasingly attractive." It is defined through the marginal contribution of a consumer to the imbalance cost.

[3]Unlike the two-price imbalance settlement, $c(S)$ can be negative under a one-price scheme, implying imbalance can be profitable.

Definition 3 (Marginal Cost Contribution)**.** *The marginal cost contribution of consumer* i *to a coalition* $S \subseteq N \setminus \{i\}$ *is* $M_i(S) = c(S \cup \{i\}) - c(S)$*, the change in coalition imbalance cost when* i *joins* S*. For the grand coalition, we write* $M_i = M_i(N \setminus \{i\})$*.*

Definition 4 (Concavity)**.** *A game is concave if*

$$c(S_1 \cup S_2) + c(S_1 \cap S_2) \leq c(S_1) + c(S_2), \quad \forall S_1, S_2 \subseteq N. \tag{5}$$

This condition has a direct reading in terms of marginal cost contributions. Taking $S_1 = A \cup \{i\}$ and $S_2 = B$ with $A \subseteq B$ and $i \notin B$, chosen here only to expose this reading and not as a restriction, the definition rearranges to $M_i(B) \leq M_i(A)$. Concavity therefore requires that a consumer's marginal contribution to imbalance cost does not increase as the coalition it joins grows, so that adding a consumer to a larger coalition saves at least as much as adding it to a smaller one. It is a desirable property because it guarantees that a *stable* allocation exists, a notion made precise later.

The imbalance netting game in our setting, though sub-additive, is *not* concave. As a minimal example, suppose the power system is short, and let the smaller coalition A be short too with total imbalance $\Delta_A = -2$ MW, and let the larger coalition $B \supseteq A$ be long overall with $\Delta_B = +1$ MW. A long consumer i with $\Delta_i = +1$ MW offsets part of A's penalized shortfall, lowering A's imbalance cost and giving a strictly negative $M_i(A)$, as $c(A \cup \{i\}) < c(A)$. However, i only deepens B's already-helping long position, leaving B's imbalance cost at zero and giving $M_i(B) = 0$. Hence, $M_i(B) > M_i(A)$ despite $A \subseteq B$, so the game is not concave. This distinction is central to the paper: stability cannot be guaranteed under non-concavity and must instead be checked for each allocation mechanism, which is the subject of Table I and Section V.

The second question is how to allocate the coalition cost among the consumers. An allocation vector $\mathbf{x}^S = \{x_i^S \,|\, i \in S\}$ assigns a cost share x_i^S to each consumer i in coalition S. In particular, the *grand-coalition allocation* is denoted by $\mathbf{x}^N = \{x_i^N \,|\, i \in N\}$. For notational simplicity, we omit the superscript N when referring to the grand-coalition allocation and write $\mathbf{x} = \{x_i \,|\, i \in N\}$. For an allocation to hold a coalition together, two minimal conditions must hold: no consumer should do worse than on its own, and the full coalition cost should be distributed.

Definition 5 (Individual Rationality)**.** *An allocation vector* x^S *is individually rational if* $x_i^S \leq c(\{i\})$ *for all* $i \in S$*.*

Definition 6 (Collective Rationality)**.** *An allocation vector* x^S *is collectively rational if* $\sum_{i\in S} x_i^S = c(S)$*.*

Definition 7 (Imputation)**.** *An allocation vector is an imputation if it is both individually and collectively rational.*

Individual and collective rationality are necessary but not sufficient: an imputation can still leave a subgroup of consumers better off on their own, in which case the grand coalition is *unstable*. To detect this, we measure how much a coalition could gain by breaking away.

Definition 8 (Excess)**.** *The excess of a coalition $S \subseteq N$ under the grand-coalition allocation $\mathbf{x}$ is*

$$\epsilon(S, \mathbf{x}) = c(S) - \sum_{i \in S} x_i. \tag{6}$$

A negative excess means that a coalition can reduce its total charge relative to what it is charged under the grand-coalition allocation $\mathbf{x}$, and therefore has an incentive to deviate. Requiring the excess of every coalition to be non-negative gives rise to the central notion of *stability* used throughout this paper.

Definition 9 (Core)**.** *The core is the set of imputations with $\epsilon(S, \mathbf{x}) \geq 0$ for all $S \subseteq N$:*

$$C = \left\{ \mathbf{x} \,\middle|\, \sum_{i \in N} x_i = c(N),\ \epsilon(S, \mathbf{x}) \geq 0,\ \forall S \subseteq N \right\}. \tag{7}$$

Intuitively, the core is the feasible region of allocations that distribute the entire grand-coalition value and under which no coalition, from a single consumer to any subgroup, can do better by breaking away. An allocation in the core is therefore stable, since no group has an incentive to leave; we adopt such stability as our notion of a well-behaved allocation. The core can be empty, and even when it is non-empty, computing an allocation inside it is non-trivial. This is the task of a *core-selecting mechanism*, or *allocation mechanism*, which maps the coalitions' characteristic functions to a single allocation. The six mechanisms presented in the next section are alternative such mappings.

III. Imbalance cost allocation mechanisms

An allocation mechanism is the third step of the pipeline in Fig. 2: it maps the characteristic function to a single allocation $\mathbf{x}$ that divides the grand-coalition cost among consumers. We study six allocation mechanisms: the Shapley value, the marginal cost contribution mechanism, the VCG mechanism, the nucleolus, the marginal price allocation mechanism, and the Gately point. Hereafter, allocations produced by these mechanisms are denoted using superscripts. For example, the Shapley value allocation is denoted by $\mathbf{x}^{\mathrm{SV}} = \{x_i^{\mathrm{SV}} \,|\, i \in N\}$.

A. Shapley Value

The Shapley value [16] assigns each consumer i a weighted average of its marginal cost contribution $M_i(S) = c(S \cup \{i\}) - c(S)$ over all coalitions S not containing i:

$$x_i^{\mathrm{SV}} = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!}\, M_i(S), \quad \forall i \in N, \tag{8}$$

where $|S|$ and $|N|$ denote the number of consumers in coalition S and the grand coalition N, respectively. The Shapley value is the unique allocation satisfying the standard fairness axioms of cooperative game theory [16]. For concave games, the Shapley value belongs to the core [17]; however, the imbalance netting game is non-concave, so this guarantee does not apply here. Moreover, computing the Shapley value requires evaluating the imbalance cost of every coalition, that is, all $2^{|N|}$ coalition costs.

B. Marginal Cost Contribution (MCC)

This mechanism charges each consumer i its marginal cost contribution to the grand coalition, as defined in Definition 3:

$$x_i^{\mathrm{MCC}} = M_i = c(N) - c(N \setminus \{i\}), \qquad \forall i \in N, \tag{9}$$

that is, the additional imbalance cost incurred by the grand coalition due to the inclusion of consumer i. Computing the MCC allocation requires evaluating the imbalance costs of only the $|N|$ leave-one-out coalitions and the grand coalition, for a total of $|N| + 1$, coalition cost evaluations. This is computationally much more efficient than the Shapley value, which requires evaluating all $2^{|N|}$ coalition costs.

C. VCG

The VCG mechanism [18]–[20] charges each consumer i the additional imbalance cost that its participation imposes on the *remaining* consumers in the grand coalition:

$$x_i^{\mathrm{VCG}} = \sum_{t \in \mathcal{T}} \lambda_{t,N}\, \Delta_{t,N\setminus\{i\}} - c(N \setminus \{i\}), \quad \forall i \in N, \tag{10a}$$

where $c(N \setminus \{i\}) = \sum_{t \in \mathcal{T}} \lambda_{t,N\setminus\{i\}}\, \Delta_{t,N\setminus\{i\}}$ is the imbalance cost incurred by the remaining consumers when consumer i is absent. The first term represents the imbalance cost that the remaining consumers would incur if their imbalance $\Delta_{t,N\setminus\{i\}}$ were instead settled at the grand-coalition imbalance price spread $\lambda_{t,N}$. Equivalently,

$$x_i^{\mathrm{VCG}} = \sum_{t \in \mathcal{T}} \big(\lambda_{t,N} - \lambda_{t,N\setminus\{i\}}\big)\, \Delta_{t,N\setminus\{i\}}, \quad \forall i \in N, \tag{10b}$$

showing that consumer i is charged only in hours where its participation changes the imbalance price spread faced by the remaining consumers, that is, where $\lambda_{t,N} \neq \lambda_{t,N\setminus\{i\}}$.[4] Under the price-taking assumption of the facilitator (Remark 3), this occurs only when the participation of consumer i changes the sign of the portfolio's imbalance; otherwise, consumer i is not charged. Consequently, the VCG mechanism charges only those consumers whose participation changes the portfolio's imbalance direction. In practice, this typically means that most consumers are not charged, leaving the facilitator with a budget deficit, which is a well-known drawback of the VCG mechanism. As with the MCC mechanism, the VCG mechanism requires evaluating only the grand coalition and the $|N|$ leave-one-out coalitions, for a total of $|N|+1$ coalition cost evaluations.

D. Nucleolus

The Nucleolus [21] selects the imputation that lexicographically maximizes the vector of coalition excesses $\epsilon(S, \mathbf{x})$ sorted in non-decreasing order. With the excess definition in (6), the smallest excess corresponds to the coalition with the strongest incentive to deviate. The Nucleolus therefore first makes this smallest excess as large as possible, then the second smallest, and so on. It has no closed form and is computed through a sequence of linear programs involving constraints for all coalitions. Since all $2^{|N|}$ coalition values are required as

[4]This pricing choice for the absent coalition Ni is a modeling decision, not forced by the game; it is what makes Proposition IV.1's closed form possible.

constraints, the method becomes computationally challenging for large portfolios. Whenever the core is non-empty, the Nucleolus lies in it, so we use it as a stability benchmark.

E. Marginal Price

The marginal price mechanism [12], a uniform-price scheme related to the shadow price imputation of [13], charges each consumer its own imbalance at the grand-coalition spread each hour:

$$x_i^{\mathrm{MP}} = \sum_{t\in\mathcal{T}} \lambda_{t,N}\, \Delta_{t,i}, \qquad \forall i \in N. \tag{11}$$

Unlike the two-price imbalance settlement, where each consumer's imbalance is settled according to whether it helps or harms the power system, this mechanism settles every consumer's imbalance at the single grand-coalition imbalance price spread, $\lambda_{t,N}$, regardless of the consumer's own imbalance direction. It therefore constitutes an internal one-price settlement scheme. In time periods where the facilitator's portfolio helps the power system, $\lambda_{t,N} = 0$, so no consumer is charged for that period. Since the imbalance cost is computed only for the grand coalition, this mechanism is computationally the most efficient among those considered.

F. Gately Point

The Gately point allocates according to each consumer's *propensity to disrupt*, seeking to balance departure incentives across consumers. In a cost-minimization setting, the propensity to disrupt of consumer i is defined as:

$$d_i(\mathbf{x}^{\mathrm{GP}}) = \frac{\sum_{j\neq i} x_j^{\mathrm{GP}} - c(N\setminus\{i\})}{x_i^{\mathrm{GP}} - c(\{i\})}, \tag{12a}$$

where the numerator is the negative of the cost savings others gain from i's participation, and the denominator is the negative of the cost savings i gains from being in the coalition. Equivalently, d_i is the ratio of cost savings for the others to cost savings for i itself. When both benefit from aggregation, this ratio is positive ($d_i > 0$), indicating that i is a valuable contributor, i.e., the others benefit substantially from including i relative to i's own benefit.

The Gately point is the imputation that minimizes the largest propensity to disrupt across consumers,

$$\mathbf{x}^{\mathrm{GP}} = \arg\min_{\mathbf{x}^{\mathrm{GP}}\in\mathcal{X}} \max_{i\in N} d_i(\mathbf{x}^{\mathrm{GP}}), \tag{12b}$$

where $\mathcal{X}$ is the set of imputations of N. This minimization balances the incentives to depart: by reducing the maximum propensity, the allocation makes no consumer too eager to leave. It admits a closed form [22] at the allocation where all propensities to disrupt are equal, with common value

$$d^* = \frac{\sum_{j\in N} M_j - c(N)}{c(N) - \sum_{j\in N} c(\{j\})}, \tag{12c}$$

which, substituted into (12a), gives each consumer's allocation

$$x_i^{\mathrm{GP}} = \frac{d^*\, c(\{i\}) + M_i}{d^* + 1}, \qquad \forall i \in N. \tag{12d}$$

The Gately point requires $2|N| + 1$ coalition costs, namely $c(N)$, the $|N|$ leave-one-out costs $c(N\setminus\{i\})$, and the $|N|$ singletons $c(\{i\})$, so it scales linearly with portfolio size.

G. Properties and Comparison

We compare mechanisms on four properties [7], [14]. *Computational complexity* is the number of coalition costs $c(S)$ that must be evaluated, together with the effort to form the allocation from them, as a function of portfolio size. *Budget balance* requires the consumer charges to sum exactly to $c(N)$, leaving the facilitator with neither surplus nor deficit. *Group rationality* requires every coalition to be at least as well off in the grand coalition as by splitting off, so that no subgroup has an incentive to leave. *Additivity* requires the allocation of the whole game to equal the sum of the allocations of its per-period, e.g., hourly, subgames. *Incentive compatibility* is not compared: by Remark 2 all forecasts are made centrally by the facilitator, so consumers cannot misreport, and the property does not arise.

Table I summarizes how the six mechanisms compare across these properties. The Shapley value and the Nucleolus both require all $2^{|N|}$ coalition costs, and the Nucleolus additionally solves up to $2^{|N|} - 1$ linear programs, which is intractable as the portfolio grows. The Shapley value further loses its core guarantee here, since the game is non-concave. VCG is cheap to compute but is never budget balanced, except in the degenerate case $c(N) = 0$, for this game and always favors the consumers. This shortfall relative to MCC is not incidental, it follows exactly from the additive structure of the game. Since imbalances are additive across consumers, $\Delta_{t,N\setminus\{i\}} = \Delta_{t,N} - \Delta_{t,i}$ for every $t \in \mathcal{T}$, and substituting into (10a) gives

$$x_i^{VCG} = x_i^{MCC} - x_i^{MP}, \qquad \forall i \in N, \tag{13}$$

where x_i^{MCC} is the marginal cost contribution (9) and x_i^{MP} is the marginal price allocation (11). Because the marginal price allocation is budget balanced ($\sum_{i\in N} x_i^{MP} = c(N)$), summing (13) over all consumers gives

$$\sum_{i\in N} x_i^{VCG} = \sum_{i\in N} x_i^{MCC} - c(N). \tag{14}$$

Consequently, whenever MCC itself falls short of budget balance (Theorem IV.4), VCG's shortfall is larger by exactly $c(N)$, consistent with the observation above that VCG typically charges only pivotal consumers and collects markedly less than $c(N)$. This identity is a consequence of the linear, per-hour price-spread structure of the imbalance netting game (4) and does not hold for VCG, MCC, and marginal price allocations in cooperative games in general. Two properties require results specific to this game, namely the budget balance of MCC and whether the Gately point exists at all, which we establish analytically in Section IV before the case study in Section V.

IV. ANALYTICAL RESULTS

This section presents two analytical results on the imbalance netting game. The first is a necessary and sufficient condition for the budget balance of MCC, which underpins the corresponding entry in Table I. The second is a condition determining when the Gately point exists at all, since the Gately point is not defined for every game. Both results are our contributions.

TABLE I: Studied mechanisms and their properties. Complexity is the number of coalition costs $c(S)$ to evaluate, as a function of portfolio size $|N|$. Where a property is established specifically for the imbalance netting game rather than following from the mechanism's general definition, the supporting result is cited alongside the entry.

Mechanism	Complexity	Budget balanced	Group rational	Additive
Shapley value [16]	$2^{\|N\|}$	✓	–[a]	✓
MCC[f]	$\|N\|+1$	–[b]	✓ (Lem. A.1)	✓ (Lem. ??)
VCG [18]–[20]	$\|N\|+1$	–[c]	✓ (Cor. A.2)	✓ (Cor. A.3)
Nucleolus [21]	$2^{\|N\|}$	✓	✓	–
Marginal price [12], [13]	1	✓	✓[d]	✓
Gately point [22], [23]	$2\|N\|+1$	✓	–[e]	–

[a] Only guaranteed to be in the core for concave games [17].
[b] Budget balanced if and only if (17) holds (Theorem IV.4); otherwise the facilitator is left in deficit.
[c]
[c] Never budget balanced for this game: the deficit always favors consumers; see (14) for the exact relationship to MCC.
[d] Established for this class of games in [12].
[e] No formal proof; group rationality empirically observed.
[f] MCC is a marginal-contribution rule; its complexity and properties are derived directly rather than cited to prior work.

A. Budget Balance of MCC

MCC is generally not budget balanced, and whether the shortfall favors the participants or the mechanism operator depends on the setting. We show that for the imbalance netting game it always favors the consumers: the charges collected fall short of the imbalance cost, leaving the facilitator in deficit.

The MCC allocation charges each consumer its marginal contribution to the imbalance cost. Budget balance therefore holds exactly when the consumers' marginal cost contributions sum to the grand-coalition cost $c(N)$, a property we call *additivity of marginal contributions at the top level* (AMC).

Definition 10 (AMC). *A game exhibits AMC if the grand-coalition cost equals the sum of the consumers' marginal cost contributions:*

$$c(N) = \sum_{i \in N} M_i. \tag{15}$$

We first show that MCC is additive over subgames. Let $\mathrm{MCC}(c) = \left(x_1^{\mathrm{MCC}}, \ldots, x_{|N|}^{\mathrm{MCC}}\right)$ denote the allocation vector that MCC assigns in a game c, with $x_i^{\mathrm{MCC}} = c(N) - c(N \backslash \{i\})$.

Lemma IV.1. *Write the game as a sum of per-settlement-period subgames, $c = \sum_{t \in \mathcal{T}} c_t$, where c_t is the game for imbalance realization t.*[5] *MCC is additive over this decomposition:*

$$\mathrm{MCC}(c) = \sum_{t \in \mathcal{T}} \mathrm{MCC}(c_t), \tag{16}$$

where the sum is taken componentwise over consumers.

Proof. See Appendix A. □

The additive decomposition lets budget balance be checked one hour at a time and then summed over the horizon. We first give a sufficient condition.

[5] This decomposition relies on the game being temporally separable: from (4), $c(S) = \sum_{t \in \mathcal{T}} \lambda_{t,S} \Delta_{t,S}$ is a sum of independent per-hour terms, so each hour forms its own subgame. Intertemporal coupling, such as storage carrying state of charge across hours, would prevent this decomposition and the additivity below would no longer apply.

Lemma IV.2. *AMC, and hence budget balance of MCC, holds if*

$$\lambda_{t,N} = \lambda_{t,N \backslash \{i\}}, \qquad \forall i \in N,\ \forall t \in \mathcal{T}, \tag{17}$$

that is, if removing any single consumer i leaves the grand-coalition price spread $\lambda_{t,N}$ of (3a) *unchanged in every hour.*

Proof. See Appendix B. □

The condition can fail, and when it does the failure is one-directional.

Lemma IV.3. *If* (17) *does not hold, MCC returns a non-budget-balanced allocation that always leaves the facilitator in budget deficit.*

Proof. See Appendix C. □

Together, Lemmas IV.2 and IV.3 characterize budget balance for MCC.

Theorem IV.4. *The imbalance netting game under the MCC mechanism is budget balanced if and only if* (17) *holds. Otherwise, the consumer charges do not cover the imbalance cost the facilitator bears, leaving it in deficit:*

$$\sum_{i \in N} x_i^{\mathrm{MCC}} < c(N). \tag{18}$$

Proof. Lemma IV.2 gives budget balance under (17), and Lemma IV.3 gives the deficit when it fails. □

B. Existence and Uniqueness of the Gately Point

The Gately point is not defined for all cooperative games, as it requires the game to be *essential*, meaning that cooperation creates a genuine cost saving for at least one consumer relative to their standalone imbalance cost. This section establishes when the Gately point exists for the imbalance netting game and provides an alternative allocation mechanism for cases where it does not.

Definition 11 (Well-defined Gately point). *The Gately point is said to be well-defined for a cooperative game if the propensity-to-disrupt calculation in* (12a) *yields a unique solution where all propensities are equal as in* (12c).

Lemma IV.5. *For every $i \in N$, $c(\{i\}) \geq M_i$. Moreover, if the game is essential, i.e.*

$$c(N) < \sum_{j \in N} c(\{j\}),$$

then $c(\{i\}) > M_i$ for at least one $i \in N$.

Proof. See Appendix E. □

Theorem IV.6. *The Gately point is well-defined for the imbalance netting game if and only if aggregation creates a strict imbalance cost saving:*

$$c(N) < \sum_{i \in N} c(\{i\}). \tag{19a}$$

if and only if there exists at least one hour $t \in T$ such that

$$\left| \sum_{i \in N} \Delta_{t,i} \right| < \sum_{i \in N} |\Delta_{t,i}|. \tag{19b}$$

Moreover, whenever (19a)–(19b) *hold, the Gately point is the unique imputation given by* (12c)–(12d).

Proof. See Appendix E. □

Corollary IV.7. *If instead consumers' imbalances are perfectly aligned in every hour $t \in T$, i.e.*

$$\left|\sum_{i\in N} \Delta_{t,i}\right| = \sum_{i\in N} |\Delta_{t,i}|, \qquad \forall t \in T,$$

the game is inessential and the Gately point is not defined. In this case, the unique core allocation is

$$x_i^{\mathrm{GP}} = c(\{i\}), \qquad \forall i \in N, \tag{20}$$

which charges each consumer its standalone imbalance cost.

Proof. See Appendix E. □

V. Results

A. Input Data

The theoretical findings are applied to a case study of a portfolio of 19 consumers each having a PPA with a photovoltaic (PV) plant through the facilitator. Consumption data consists of historic measurements of hourly consumption data for each of the 19 consumers. Price, PV production, and PV production forecast data are provided through *Energinets* Energi Data Service [24]. All data is in hourly resolution.
Each consumer has a PPA of an individually negotiated percentage of the total PV production. As all consumers in this case study have PPA's with the same PV plant, uncertainty caused by production is not subject to netting effects. We focus our study on the year 2024, containing 366 days each with 24 hours, resulting in $T = 24 \cdot 366$ settlement periods.
The case study consists of a varied set of consumers with distinct demand curves and large variations in total demand and PV coverage, as seen in Fig. 3. This heterogenous portfolio provides a background for assessing the gains and limitations of cooperative allocation.

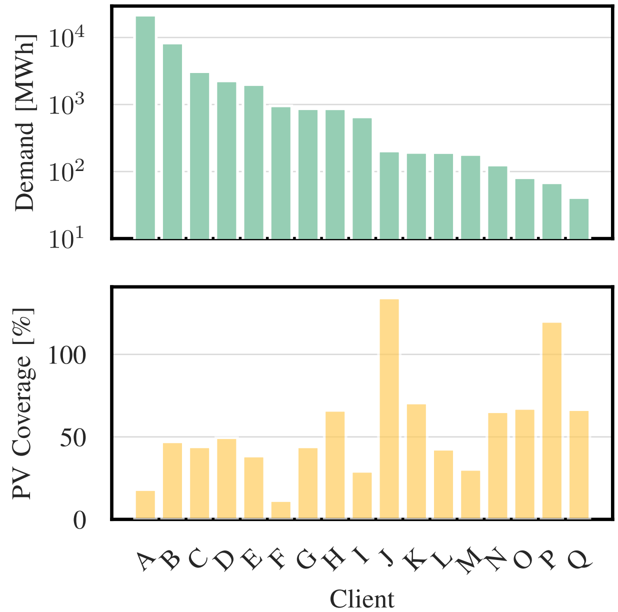

Fig. 3: Bar chart of total PV coverage, $PVCover_i = \frac{\sum_{t\in T} PVProd_i}{\sum_{t\in T} Demand_i}$, and demand of consumers in the case study period

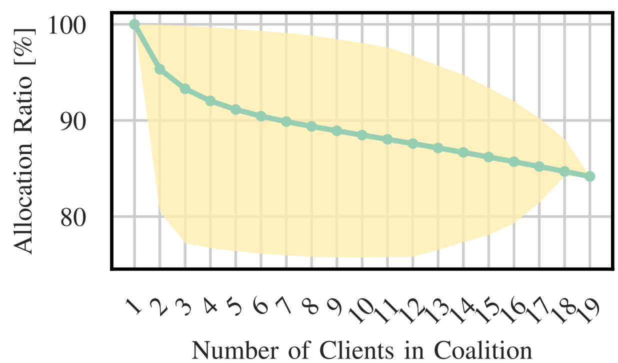

Fig. 4: Relative allocated cost by coalition size, for each coalition size n, computed as the ratio of a coalition's characteristic function to the sum of singleton costs in the coalition, unweighted average (circles) and ranged (shaded area) over all coalitions of size n.

B. Savings

To motivate the application of cooperative game theory to imbalance netting within portfolios, the gains from creating coalitions are calculated in Fig. 4. There are clear gains from aggregation in this case, as aggregated cost decreases as coalition size increases. It is also clear that some coalitions are better than others, with the most well balanced coalitions in the year having a cost less than 80%, significantly lower than the average. The challenge is then how to divide this reduction in cost among consumers in a stable manner. Another perspective to understand how well a consumer works in a portfolio is to look at their individual allocation ratios. This shows how big the reduction in cost for the consumer is when joining the grand coalition, and gives an indication on whether a consumer is allocated a low cost because they by themselves have a low imbalance, or because they are especially useful in the portfolio. The results are shown in Fig. 5. Any consumer with a allocation ratio lower than 100% will prefer to be in the grand coalition, compared to alone. We see that applying a flat rate to all consumers will make some consumers pay higher fees than their singleton cost, making it highly unstable. We also see that the allocation ratio of consumer A is very close to 100 for all allocation mechanisms, indicating small gains for aggregation, while other consumers see higher gains.

C. Stability

In table II, we see the maximum excess of consumers. This will be the maximum utility gain of a coalition possible by splitting off from the grand coalition. All allocation

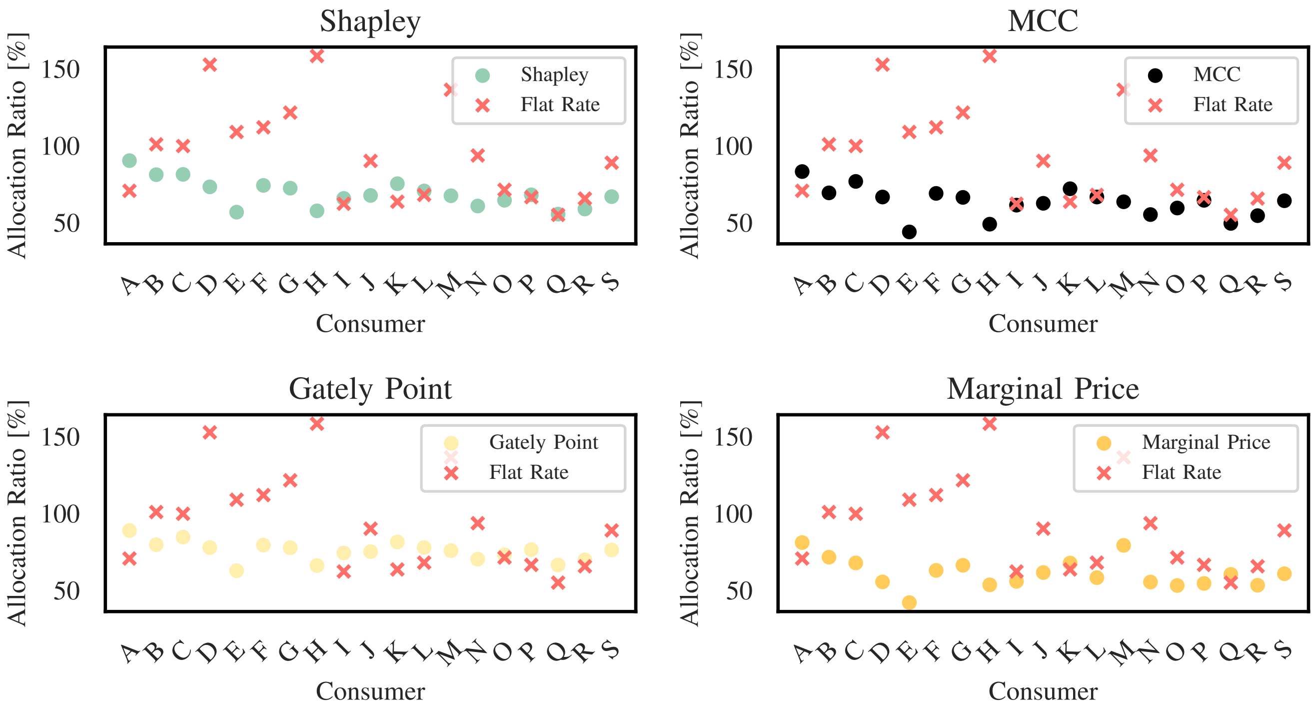

Fig. 5: Plot of ratio between allocated cost and singleton cost.

mechanism are stable in practice, except for the flat rate allocation. Notably, the marginal price allocation is the least stable of the mechanisms, although it and the nucleolus are the only coalitions proven to be stable for this game. The Nucleolus will return the most stable allocation possible, providing a benchmark for the other allocations. VCG is more stable than the nucleolus, as a result of VCG's lacking budget balance, otherwise the Gately Point gives the closest stability to the Nucleolus.

D. Business model

The business model is an important part of individualising costs. One of the inherent issues with individual pricing, especially ex-post, is that it reflects the uncertainty back onto the consumer, giving them part of the financial responsibility for the imbalances. This shifts the BRP role of the facilitator from being a form of insurance to providing a service of minimizing imbalance costs on the behalf of consumers.

This shift in the role of the facilitator is similar to how an electricity retailer can have a flat or variable price structure. Costs can either be calculated based on an expected cost with a premium, or directly reflect the costs of the consumers consumption. In practice, the benefits of individual pricing would be to incentivize large scale customers to quickly report unexpected events, such as outages, assisting the facilitator in lowering imbalance, and could be extended to reward flexible consumers for reducing imbalances within the portfolio, given the proper signals.

A fully individualized model however, does not directly incentivize the facilitator to work on reducing imbalances, only indirectly through competitiveness on the market, as all imbalance costs are forwarded to the consumer. Therefore, the solution can lie in a partial individualization, partly individualizing costs for fairness and consumer responsiveness, and partly socializing them for facilitator responsibility.

The cost structure for a consumer could be described by the following formula:

$$\mathbf{C}_{i,m} = \sum_{t\in T_m} k_t \hat{p}^D_{i,t} + \left(1-\alpha^{ind}\right)\hat{\lambda}^{soc}_m \cdot \sum_{t\in T_m} \hat{p}^D_{i,t} + \alpha^{ind}\left(\hat{\lambda}^{ind}_{i,m} \cdot \sum_{t\in T_m} \hat{p}^D_{i,t} + Q_{i,m-1}\right), \quad (21)$$

where $\mathbf{C}_{i,m}$ is the consumers cost for a month, k_t is the agreed cost per unit of energy that covers all other costs than imbalance costs e.g. spot price and overhead, $\alpha^{ind} \in [0,1]$ indicates the degree of individualization, $\hat{\lambda}^{soc}_m$ is an expected socialized cost per MWh for all consumers for the month, $\hat{\lambda}^{ind}_{i,m}$ is an expected individualized cost per MWh for the month, $\hat{p}^D_{i,t}$ is the consumers expected net consumption per imbalance settlement period and $Q_{i,m-1}$ is a reconciliation payment applied ex-post that constitutes the difference between the expected individual cost for the previous month $\hat{\lambda}^{ind}_i \cdot \sum_{t\in T_{m-1}} \left(\hat{p}^D_{i,t}\right)$ and the realized allocated individual cost for the previous month $x^N_{i,m-1}$:

$$Q_{i,m} = x^N_{i,m} - \hat{\lambda}^{ind}_i \cdot \sum_{t\in T_m} \left(\hat{p}^D_{i,t}\right). \quad (22)$$

Clients	Shapley	VCG	Marginal Price	Gately Point	MCC	Flat Rate	Nucleolus
19	-1.9	-12789	-28.4	-8.7	-25.9	9768	–

TABLE II: Maximum excess of allocations, negative excess provides stability. MCC and VCG are not budget balanced.

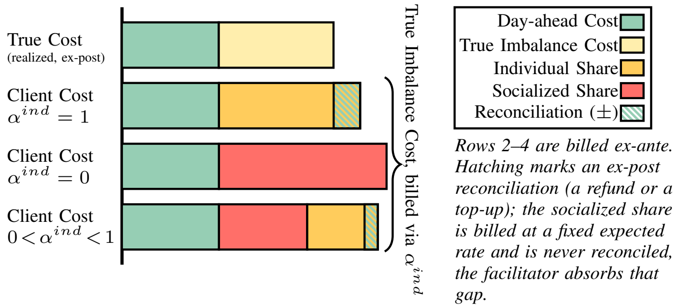

Fig. 6: A consumer's billed cost across individualization grades α^{ind} (schematic; not to scale). Top: realized ex-post cost. Below: ex-ante billed cost under full individualization, full socialization, and a mixed case. Only the individualized share is reconciled (hatched, Eq. 22); the socialized share is billed at a fixed rate and never reconciled, so the facilitator absorbs its ex-post deviation.

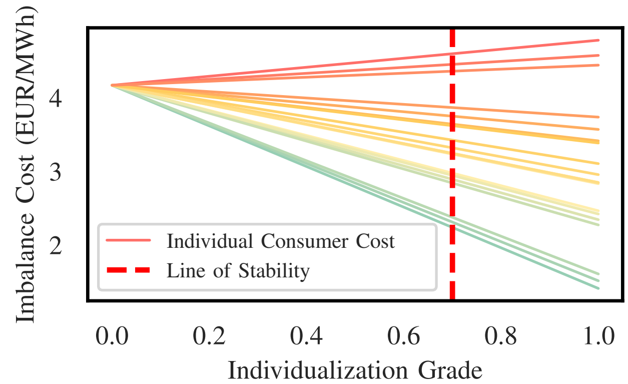

Fig. 7: The cost assigned by the marginal price allocation to consumers dependent on cost individualisation

A consumers total cost includes: (1) a base cost at hourly price k_t; (2) a socialized share billed ex-ante at a common expected rate $\hat{\lambda}^{soc}_m$ and never reconciled, so the facilitator absorbs any gap to the realized cost; and (3) an individualized share billed at $\hat{\lambda}^{ind}_{i,m}$ and reconciled ex-post via $Q_{i,m}$ to match the consumers true cost $x^N_{i,m}$. This is illustrated in Fig. 6: full individualization ($\alpha^{ind} = 1$) reconciles fully (a refund or top-up); full socialization ($\alpha^{ind} = 0$) never reconciles; a mixed case reconciles only the individualized portion. As described in section V-D, it might be preferable not to reflect the entire individualised cost to consumers. In Fig. 7 it is shown how the cost allocated to consumers changes based on the grade of individualisation, with an individualisation grade of 0 being fully socialised and 1 being fully individualised.

The allocation is group rational above an individualisation grade of 0.85. Below this, some may can split off in order to increase their utility (by joining a fully individualised sub-coalition). The choice of individualisation grade is a contract parameter that is a trade-off between fairness, risk, and incentivising consumers and facilitator to reduce balancing costs.

VI. Conclusion and Future Work

This work studied the allocation of imbalance costs within the portfolio of a Balance Responsible Party using a cooperative game-theoretic framework. Several allocation mechanisms were analyzed, ranging from marginal pricebased rules to established cooperative solutions such as the Gately Point, VCG, and the Nucleolus. While cooperative mechanisms provide appealing fairness and stability properties, their practical implementation is constrained by computational complexity and budget-balance considerations. Consequently, no single allocation mechanism is strictly superior to the others.

The numerical analysis demonstrates that aggregation consistently reduces total imbalance costs, independent of the allocation mechanism applied. What distinguishes different mechanisms is therefore not the operational outcome of the portfolio, but how the gains from aggregation are redistributed among consumers. In particular, substantial heterogeneity in netting contributions motivates moving beyond socialized flat-rate pricing toward individualized or partially individualized schemes.

From a practical perspective, fully individualized pricing shifts imbalance risk entirely to consumers, whereas fully socialized pricing neglects consumer heterogeneity. A partially individualized cost allocation emerges as a compromise that allows facilitators to balance fairness, risk sharing, and incentive alignment. The preferred degree of individualization thus depends on contractual preferences.

Finally, the results rely on an asymmetric ttwo-price imbalance settlement, under which aggregation yields direct cost savings. Under symmetric imbalance pricing, aggregation primarily reduces cost volatility rather than expected cost, and the implications for allocation mechanisms should be reassessed. Future work should therefore consider symmetric pricing schemes, strategic behaviour of consumers, and the inclusion of flexible resources such as storage, which may alter both the netting potential and the stability of cooperative allocations.

Appendix

A. Additivity of MCC

Proof. The imbalance netting game decomposes over the horizon: defining the hourly subgame $c_t(S) = \lambda_{t,S}\Delta_{t,S}$, we have $c(S) = \sum_{t\in\mathcal{T}} c_t(S)$ for every $S \subseteq N$ by (4). The payment to consumer i is its marginal cost contribution. Substituting the decomposition,

$$\begin{aligned} x_i^{\mathrm{MCC}} &= c(N) - c(N \setminus \{i\}) \\ &= \sum_{t \in \mathcal{T}} c_t(N) - \sum_{t \in \mathcal{T}} c_t(N \setminus \{i\}) \\ &= \sum_{t \in \mathcal{T}} \big[c_t(N) - c_t(N \setminus \{i\})\big] = \sum_{t \in \mathcal{T}} M_{t,i}, \end{aligned} \tag{23}$$

where $M_{t,i} = c_t(N) - c_t(N \setminus \{i\})$ is consumer i's marginal cost contribution in hour t. Thus, each consumer's MCC charge is the sum of its hourly marginal cost contributions, i.e., $\mathrm{MCC}(c) = \sum_{t \in \mathcal{T}} \mathrm{MCC}(c_t)$ componentwise, proving Lemma IV.1. □

B. Conditions for Budget Balance of MCC

Proof. Consider a single-hour subgame c_t of c. By (4), its characteristic function is

$$c_t(N) = \lambda_{t,N}\, \Delta_{t,N} = \lambda_{t,N} \sum_{i \in N} \Delta_{t,i}. \tag{24a}$$

Assume that in this hour, the price spread is unchanged when any single consumer leaves the coalition,

$$\lambda_{t,N} = \lambda_{t,N \setminus \{i\}}, \qquad \forall i \in N. \tag{24b}$$

Under this condition, $c_t(N \setminus \{i\}) = \lambda_{t,N\setminus\{i\}} \Delta_{t,N\setminus\{i\}} = \lambda_{t,N} \Delta_{t,N\setminus\{i\}}$, so the marginal contribution of consumer i is

$$\begin{aligned} M_{t,i} &= c_t(N) - c_t(N \setminus \{i\}) \\ &= \lambda_{t,N}\Delta_{t,N} - \lambda_{t,N}\Delta_{t,N\setminus\{i\}} = \lambda_{t,N}\Delta_{t,i}, \end{aligned} \tag{24c}$$

using $\Delta_{t,N} - \Delta_{t,N\setminus\{i\}} = \Delta_{t,i}$. Summing over consumers,

$$\sum_{i \in N} M_{t,i} = \lambda_{t,N} \sum_{i \in N} \Delta_{t,i} = c_t(N). \tag{24d}$$

Hence, AMC as defined in (15) holds for hour t. Since MCC allocates marginal contributions, AMC implies that the consumer charges sum to $c_t(N)$, so MCC is budget balanced in that hour. By the additivity of Lemma IV.1, budget balance in every hour gives budget balance over the whole horizon, proving Lemma IV.2. □

C. Conditions for Missing Budget Balance of MCC

Proof. Consider an hour t with

$$\delta_t = \{\, i \in N \mid \lambda_{t,N} \neq \lambda_{t,N\setminus\{i\}} \,\} \neq \emptyset. \tag{25}$$

The facilitator and every consumer within the portfolio are price-takers, so removing a consumer changes the spread $\lambda_{t,N}$ only by flipping the sign of the coalition imbalance (see Remark 3). Hence, each $i \in \delta_t$ has $\Delta_{t,i}$ sharing the sign of $\Delta_{t,N}$ with $|\Delta_{t,i}| > |\Delta_{t,N}|$, so its removal flips the coalition between helping and harming the power system. A consumer $i \notin \delta_t$ leaves the spread unchanged and satisfies $M_{t,i} = \lambda_{t,N}\Delta_{t,i}$, as in Appendix B. Two cases arise, according to the direction of the grand coalition in hour t.

Case 1: the grand coalition harms, so $c_t(N) = \lambda_{t,N}\Delta_{t,N} > 0$. Removing any $i \in \delta_t$ flips the remainder to helping, which settles at the day-ahead price, so $c_t(N \setminus \{i\}) = 0$ and thus $M_{t,i} = c_t(N)$. Summing over all consumers and using the identity $\lambda_{t,N} \sum_{i \in N} \Delta_{t,i} = c_t(N)$ to replace the non-flipping terms,

$$\begin{aligned} \sum_{i \in N} M_{t,i} &= |\delta_t|\, c_t(N) + \lambda_{t,N} \sum_{i \notin \delta_t} \Delta_{t,i} \\ &= (|\delta_t| + 1)\, c_t(N) - \lambda_{t,N} \sum_{i \in \delta_t} \Delta_{t,i}, \end{aligned} \tag{26a}$$

where $|\delta_t|$ is the number of consumers in δ_t. For each $i \in \delta_t$, $\Delta_{t,i}$ shares the sign of $\Delta_{t,N}$ and exceeds it in magnitude, so $\lambda_{t,N}\Delta_{t,i} > \lambda_{t,N}\Delta_{t,N} = c_t(N)$; hence $\lambda_{t,N} \sum_{i \in \delta_t} \Delta_{t,i} > |\delta_t|\, c_t(N)$. For each $i \in \delta_t$, $\Delta_{t,i}$ shares the sign of $\Delta_{t,N}$ and exceeds it in magnitude, so $\lambda_{t,N}\Delta_{t,i} > \lambda_{t,N}\Delta_{t,N} = c_t(N)$; hence $\lambda_{t,N} \sum_{i \in \delta_t} \Delta_{t,i} > |\delta_t|\, c_t(N)$. *This inequality $M_{t,i} < \lambda_{t,N}\Delta_{t,i}$ holds termwise for each individual $i \in \delta_t$, not merely in aggregate, a fact used directly in the proof of Lemma A.1..* Substituting into (26a),

$$\sum_{i \in N} M_{t,i} < (|\delta_t| + 1)\, c_t(N) - |\delta_t|\, c_t(N) = c_t(N). \tag{26b}$$

Case 2: the grand coalition helps, so $c_t(N) = 0$ and $\lambda_{t,N} = 0$. Every non-flipping consumer then has $M_{t,i} = \lambda_{t,N}\Delta_{t,i} = 0$, while removing any $i \in \delta_t$ flips the remainder to harming, giving $c_t(N \setminus \{i\}) > 0$ and $M_{t,i} = c_t(N) - c_t(N \setminus \{i\}) = -c_t(N \setminus \{i\})$. Therefore

$$\sum_{i \in N} M_{t,i} = -\sum_{i \in \delta_t} c_t(N \setminus \{i\}) < 0 = c_t(N). \tag{26c}$$

In both cases, $\sum_{i \in N} M_{t,i} < c_t(N)$ whenever $\delta_t \neq \emptyset$. Since MCC allocates $M_{t,i}$ to each consumer, the charges collected in such an hour sum to strictly less than the hourly imbalance cost, leaving the facilitator in budget deficit. The result extends to any game containing such an hour, so MCC is not budget balanced whenever (17) fails, and the shortfall always favors the consumers. □

D. Group Rationality of MCC and VCG

Lemma A.1 (Group rationality of MCC)**.** *For every consumer $i \in N$,*

$$x_i^{\mathrm{MCC}} \leq x_i^{\mathrm{MP}}, \tag{27}$$

where x_i^{MP} is the marginal price allocation of (11)*. Consequently, MCC is group rational: since marginal price is group rational for this class of games [12], for every coalition $S \subseteq N$,*

$$\sum_{i \in S} x_i^{\mathrm{MCC}} \leq \sum_{i \in S} x_i^{\mathrm{MP}} \leq c(S), \tag{28}$$

i.e., $\epsilon(S, x^{\mathrm{MCC}}) \geq 0$ for all $S \subseteq N$, so no coalition can lower its total charge by leaving the grand coalition under MCC.

Proof. By construction, $x_i^{\mathrm{MP}} = \sum_{t \in \mathcal{T}} \lambda_{t,N}\Delta_{t,i}$. Appendix C (proof of Lemma IV.3) shows termwise that $M_{t,i} = \lambda_{t,N}\Delta_{t,i}$ for $i \notin \delta_t$, and $M_{t,i} < \lambda_{t,N}\Delta_{t,i}$ for $i \in \delta_t$. Summing over $t \in \mathcal{T}$ gives $x_i^{\mathrm{MCC}} \leq x_i^{\mathrm{MP}}$ for every $i \in N$, proving (27). Inequality (28) then follows by summing over $i \in S$ and invoking the group rationality of marginal price [12]. □

Corollary A.2 (Group rationality of VCG)**.** *The VCG allocation is group rational.*

Proof. By (13), $x_i^{\mathrm{VCG}} = x_i^{\mathrm{MCC}} - x_i^{\mathrm{MP}}$ for all $i \in N$. Lemma A.1 gives $x_i^{\mathrm{MCC}} \leq x_i^{\mathrm{MP}}$, so $x_i^{\mathrm{VCG}} \leq 0$ for every consumer — under VCG, no consumer is ever charged a positive net amount in this game. Since $c(S) \geq 0$ for every coalition S (Section II-B), it follows immediately that for any $S \subseteq N$,

$$\sum_{i \in S} x_i^{\mathrm{VCG}} \leq 0 \leq c(S), \tag{29}$$

so $\epsilon(S, x^{\mathrm{VCG}}) \geq 0$ for all $S \subseteq N$, establishing group rationality. □

Corollary A.3 (Additivity of VCG)**.** *The VCG mechanism is additive.*

Proof. By (13), $x_i^{\mathrm{VCG}} = x_i^{\mathrm{MCC}} - x_i^{\mathrm{MP}}$ for all $i \in N$. MCC is additive (Lemma IV.1) and marginal price is additive by construction, so their difference is additive over the same per-period decomposition, proving the claim. □

E. Existence, Well-Definedness, and Uniqueness of the Gately Point

Proof. By (12c), the Gately point exists precisely when $c(N) - \sum_{j \in N} c(\{j\}) \neq 0$. Since the game is sub-additive through Definition 2 we have $c(N) \leq \sum_{j \in N} c(\{j\})$; the denominator of (12c) is therefore never positive, and the Gately point exists if and only if

$$c(N) < \sum_{j \in N} c(\{j\}). \tag{30}$$

Consider an hour $t \in T$. By the two-price settlement (Section II-B), the hourly cost $c_t(S)$ of any coalition S depends only on whether S is long or short that hour, and is proportional to $(\Delta_{t,S})^+$ or $(\Delta_{t,S})^-$ accordingly, with the same proportionality constant for every S (Remark 3). Either way, $\sum_i c_t(\{i\}) \geq c_t(N)$, with equality unless the consumers' imbalances mix signs that hour, i.e. unless $\left|\sum_i \Delta_{t,i}\right| < \sum_i |\Delta_{t,i}|$; when they do, any consumer i whose imbalance opposes the portfolio's satisfies $c_t(\{i\}) + c_t(N \setminus \{i\}) > c_t(N)$, and since every other hour contributes a non-negative term to the same sum, $c(\{i\}) + c(N \setminus \{i\}) > c(N)$ overall, i.e. $c(\{i\}) > M_i$ strictly.

This, together with $c(\{i\}) \geq M_i$ for every i from subadditivity, are exactly the conditions of Staudacher and Anwander [25] guaranteeing that whenever (30) holds, the equal-propensity solution (12c) is the unique imputation.

Conversely, if $\left|\sum_i \Delta_{t,i}\right| = \sum_i |\Delta_{t,i}|$ for every $t \in T$, every hourly gap is zero, so $c(S) = \sum_{i \in S} c(\{i\})$ for all $S \subseteq N$: the game is inessential, and the unique core allocation is $x_i^{\mathrm{GP}} = c(\{i\})$ for all $i \in N$. □

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